

NOVA

IMS

Information
Management
School

Master

MDSAA

Degree Program in
Data Science and Advanced Analytics

Business Process Management

Assignment 2

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Universidade Nova de Lisboa

June, 2026

INDEX

1. Executive Summary	3
2. Introduction	4
3. ETL process	4
3.1. Data Sources.....	4
3.2. Data Loading Process	5
3.3. Data Model.....	5
3.4. Data Quality.....	5
3.5. Main Insights from ETL.....	6
4. Dashboard	6
4.1. Process Discovery.....	6
4.2. Variant Explorer.....	8
4.3. Case Explorer.....	9
4.4. Business Analysis.....	9
4.5. Satisfaction Analysis.....	10
4.6. Conformance Checking	11
4.7. Time and Efficiency Analysis	14
4.8. Claims Refusal Analysis	15
4.9. Communication Analysis	17
5. Conclusions.....	18
5.1. Summary of Findings.....	18
5.2. Main Problems Identified.....	18
5.3. Recommendations	19
5.4. Value Generated	19

1. EXECUTIVE SUMMARY

SafeDrive Insurance is a motor vehicle insurance company serving both individual and corporate customers across Portugal. Over recent years, claims volume has increased significantly while customer satisfaction has declined. To understand the causes of this situation, a process mining analysis was conducted on the website Celonis, covering 23,275 claims and 233,652 events recorded between January 2020 and October 2024.

The analysis revealed that the claims handling process is highly complex, with 5,450 distinct process variants and a 0% conformance rate to the ideal process model, exposing a severe lack of standardization across the organization. From a business perspective, operations are heavily concentrated in Lisboa, accounting for over 60% of all claims, and in Motorcycle claims (68%), which represent the largest vehicle segment. Although Glass Breaking is the most frequent claim reason, Collision/Impact claims carry the highest financial weight with an average claim amount of 2,470.23€, more than three times higher than Glass Breaking at 715.06€.

Processing times vary enormously, averaging 32.8 days overall but reaching up to 99 days in Braga and 67 days for Collision/Impact claims. Customer satisfaction averages at 3.18 out of 5, with nearly 10% of customers giving the lowest score of 1. A critical communication failure was also identified: despite high automation, 11.6% of customers were forced to call to request information, and customers sent more emails to the company than they received, strongly suggesting systemic shortcomings in proactive communication.

To address these issues, SafeDrive Insurance should standardize its process, investigate severe regional delays, create a fast-track for Collision/Impact claims, improve proactive communication, and invest in staff training. This analysis provides a clear data-driven foundation to guide these improvements and generate measurable value for the organization.

2. INTRODUCTION

SafeDrive Insurance is a company specialized in motor vehicle insurance, providing coverage for private and commercial vehicles, across Portugal and serving both individual and corporate customers.

Over the past years, the number of claims submitted by policyholders has increased significantly. While the company has managed to process higher volumes of claims, the customer satisfaction indicators have declined. Many customers have reported that the company is taking too much processing time, have been giving unclear communication and delaying compensation payments. The management team suspects that the claims process may be inefficient and contributing to these issues.

So, with that in mind, the goal set for this project is to analyze the AS-IS claims handling process, using process mining techniques on the Celonis Website. More specifically, the project aims to identify inefficiencies and bottlenecks in the claims process, understanding what factors drive the customer dissatisfaction, detect deviations from ideal processes and provide data-driven solutions and recommendations to improve the process performance and the customer experience.

To support this analysis, two data tables were provided covering the period between January 2020 to October 2024. The Case Table contains contextual information about each individual claim, while the Active Table contains the full event log of all the activities performed throughout the claim process. Together these tables allow a comprehensive process mining analysis of the company's claims on handling operations.

3. ETL PROCESS

3.1. DATA SOURCES

The foundational step of the process mining initiative involved extracting and preparing two core CSV files from the operational database to be loaded into the Celonis environment.

The first file is the Case Table, which consists of 23,275 rows with exactly one row dedicated to each individual claim, housing all relevant contextual and master information. The second file is the Activity Table, a significantly larger dataset containing 233,652 rows that comprises the granular event log tracking all historical activity steps executed per claim.

3.2. DATA LOADING PROCESS

To begin processing this information, both CSV files were securely uploaded into a centralized Celonis Data Pool to establish a secure cloud staging area. Once the ingestion was complete, custom SQL transformations were written and executed to perform an initial exploratory analysis of the data, allowing for row-count verification and structural profile auditing.

Following this exploratory data transformation phase, a formal Data Model was constructed within the platform to establish a permanent relational link between both tables for downstream process discovery.

3.3. DATA MODEL

The architecture of the newly created Data Model enforces a strict 1-to-Many (1:N) relational database structure between the Case Table and the Activity Table. This relationship is structurally connected by joining the CLAIM_NUMBER column from the primary Case Table to the corresponding CASE_KEY field in the fact-level Activity Table. This precise join matches the operational logic of the insurance pipeline, acknowledging that while each row in the master table represents an isolated claim, any single claim can natively trigger multiple sequential activities over its operational lifecycle.

3.4. DATA QUALITY

An extensive data quality assessment indicates that the ingested database is generally clean, robust, and well-structured, with no major missing values or broken fields detected across any critical attributes.

However, the exploratory data auditing flagged a few minor anomalies, such as a small subset of cases exhibiting a negative throughput time with a minimum of -1 day, which strongly points to localized date registration or manual data entry errors.

Furthermore, incidents categorized under Theft or Robbery appear very rarely across the entire log, highlighting a potential data quality or reporting omission that requires ongoing validation.

3.5. MAIN INSIGHTS FROM ETL

Compiling the data through the completed ETL pipeline delivers a comprehensive baseline of the SafeDrive Insurance operational landscape, totalling 23,275 claims processed across 233,652 distinct events.

The system identifies 13 distinct activities within the logging pipeline which combine to generate 5.45K unique process variants, a high metric that reflects extreme process complexity and fragmentation. The regional distribution is heavily concentrated as Lisboa accounts for the vast majority of the territory with 14,131 claims, while Motorcycles emerge as the most common vehicle asset class with 15,796 claims, and Glass Breaking stands out as the most frequent claim reason with 18,141 cases.

Within the automated workflow, the Automated SMS trigger serves as the most frequent activity milestone with 28,235 total occurrences, while the final gatekeeping metrics show an exceptionally low refusal rate of just 0.67%, representing a mere 155 refused claims out of the entire organizational volume.

4. DASHBOARD

4.1. PROCESS DISCOVERY

The Process Discovery dashboard provides a visual representation of the end-to-end claims handling process at the SafeDrive Insurance, mapping the actual flow of activities as recorded in the event log across all 23,275 claims.

The main path follows a clear sequential structure, beginning with the Incident Date, moving through Register Claim, Assessment Scheduled, Assessment Date, Automated SMS, and Assessment Closed, before concluding with Claim Decision, Close Claim, and finally Make Payment. This linear flow represents the intended standard process and it's the most frequently observed sequence across the dataset.

When analysing the process through Case Frequency, the flow highlights the most common paths taken across all the claims. The majority of cases follow the standard path, with Incident Date and Register Claim appearing in all cases. However, there are some notable drops at the Assessment Schedule with 19.3K cases, Automated SMS with 17.2K and Make Payment with 15.1K, confirming that a significant portion of the claims do not complete all the expected steps.

The Activity Frequency part reveals how many times each activity was executed across the entire dataset, including repeated occurrences within the same case. On the other hand, Automated SMS stands out as the most executed activity with 28.2K occurrences despite only appearing in 17.2K, suggesting that in many claims the SMS is used multiple times. This repetition is a sign of process inefficiency and points to automation failures that force the system to resend notifications.

The average throughput time view shows that the longest wait in the entire process occurs between Incident Date and Register Claim at 8 days, and between Assessment Date and Automated SMS at 11 days. The final stage between Close Claim and Make Payment adds a further 8 days. These three gaps are the primary contributors to the overall average processing time of 32.8 days and represent the most critical areas for improvement.

The median throughput time view provides a more representative picture of the typical claim journey by removing the distortion caused by extreme outlier cases. The median processing time sits significantly below the average, confirming that the overall average is heavily inflated by a small number of severely delayed cases. This suggests that while most claims are processed within a reasonable timeframe, a problematic minority of cases is dragging the overall performance metrics down considerably.

The trimmed mean throughput time removes the most extreme outliers from both ends of the distribution, providing a balanced and robust measure of central tendency. This

view is particularly useful for identifying the true typical processing time without the distortion of exceptional cases. The trimmed mean confirms that the process performs reasonably well for the majority of claims, but that targeted intervention on the outlier cases would have a significant positive impact on overall performance metrics.

A Vehicle Type filter was also added to this page, allowing for a comparison of process behaviour across Motorcycles, Fleets, and Light Vehicles, enabling the identification of whether certain vehicle types are more prone to deviations or longer processing times.

4.2. VARIANT EXPLORER

The Variant Explorer dashboard provides a detailed breakdown of all the different execution paths across the approximately 23.3K cases in the dataset processed by the company. In total, around 5,450 distinct variants were identified, covering 100% of all the cases. This extreme number of variants is a clear indicator of severe process fragmentation and lack of standardization across the organization.

In this process there are two major paths. The first one (Variant #1) is the most common path, accounting for 36% of the cases, with an average throughput time of 28 days. This variant represents the standard path, following the complete intended sequence from Incident Date through to Make Payment without any deviations.

The second main path (Variant #2) covers 15% of the cases and is completing significantly faster at an average of only 15 days. When combined with Variant #1, just two variants already account for 51% of all cases, meaning that just over half of all claims follow one of two relatively standard paths.

For the other paths, Variant #3 accounts for 2% of the claims with an average of 19 days, while variants #4 through #10 each represent only 1% or less of total cases individually. However, several of these minority variants show dramatically longer processing times, with Variant #6 averaging 121 days, Variant #8 averaging 113 days, Variant #9 averaging 113 days, and Variant #10 averaging 112 days. These variants likely involve repeated activities, missing steps, or unexpected events such as multiple email exchanges or calls to request information, which significantly inflate processing times.

When all variants are selected simultaneously, the process map on the right becomes a highly fragmented process flow, visually demonstrating just how far the real-world process deviates from the intended linear flow. This complexity makes it very difficult to manage and monitor performance consistently and strongly supports the need for process standardization as a top priority for SafeDrive Insurance.

4.3. CASE EXPLORER

The Case Explorer page serves as a granular analytical tool, allowing for an in-depth and individual claim-level examination of the operational workflow. This capability is essential for identifying specific instances of procedural bottlenecks.

For instance, the analysis highlights severe outlier cases, such as a specific claim (Case ID 51940391) that recorded a severe outlier with a throughput time of 318 days, which serves as a prime example of extreme operational delay.

Furthermore, diving into the specific activities of individual claims reveals highly inefficient communication loops. A clear example is Case ID 52900746, which took 36 days to resolve due to a repetitive cycle of sending and receiving emails late in the process.

Despite these severe anomalies, the broader evaluation indicates that the majority of cases follow a relatively standard trajectory, typically involving around 9 distinct activities and averaging a total duration of 20 to 25 days.

4.4. BUSINESS ANALYSIS

This page provides a high-level overview of the company's claims portfolio from the period provided, where SafeDrive Insurance's processed 23,275 claims, with a total claim value of almost 25.65M€ and an average claim amount of 1,101.97€. Overall, the refusal rate was very low at 0.67%, corresponding to only 155 refused claims out of the 23,275 total.

From a geographic perspective, Lisboa is by far the dominant district, accounting for a total of 14,131 claims, which corresponds to more than 60% of all claims processed, followed by Aveiro follows with 4,059 claims and Coimbra with 1,625. The other districts don't have a significant number of claims, being Viseu, Faro and Guarda with

only 197, 227 and 229 respectively. This extreme geographic concentration suggests that this company's customers are heavily based in Lisboa's metropolitan area, which has important implications for resource allocation and staffing decisions.

In terms of vehicle type, Motorcycles represent the largest segment with 15,796 claims, followed by Fleets with 4,209 claims and Light Vehicles with 3,270 claims. The dominance of motorcycle claims is relevant for business insights, as motorcycles are more typically associated with high-risk profiles and more complex claims.

Regarding the claim reasons, Glass Breaking stands out by far as the most frequent cause, with 18,141 claims, followed by Collision/Impact with 5,131 claims and Theft or Robbery with only 3 claims. However, when analysing the average claim amount, a very significant difference is noticed, the Collision/Impact claims have an average value of 2,470.23€, more than three times higher than Glass Breaking at 715.06€ and for Theft or Robbery at 611.84€. This means that although Collision/Impact claims are less frequent, they represent a disproportionately high financial impact for the company and therefore deserve a particular attention in terms of process efficiency and handling time.

Overall, the Business Analysis reveals that SafeDrive Insurance's operations are highly concentrated in Lisboa and in motorcycle and glass breaking claims, while collision/impact claims carry highest financial weight, despite being less frequent.

4.5. SATISFACTION ANALYSIS

This page examines the customer satisfaction levels across different segments of SafeDrive Insurance's claims, using a rating scale from 1 to 5.

The overall average satisfaction score was 3.18 out of 5, which is moderate and indicates that while most of the customers are reasonably satisfied, there is a significant room for improvement. The number of claims that received a low satisfaction score of 1 or 2 was 6,547, which is almost 30% of all the claims received. On the other hand, the satisfaction score of the high score of 4 or 5 was 10,115, which is almost 44% of all claims.

Looking at the distribution of scores, the satisfaction score of 4 is the most common one, with 7,045 claims, followed by the score 3 with 6,613 claims and score 2 with

4,199 claims. The highest score has only 3,070 claims, while the lowest has 2,348 claims. The fact that 10% of the claims have been rated as the lowest score of 1 is a critical insight that deserves extra attention.

When analysing satisfaction by district, Guarda has the lowest score average at 3.07, followed by Evora with 3.09 and Porto with 3.15. On the other side, the districts with the highest satisfaction rate are Braga with a score of 3.20, Viseu and Lisboa with 3.19.

Regarding claim reasons, Theft or Robbery claims have the highest satisfaction score at 3.33, followed by Glass Breaking at 3.19 and Collision/ Impact at 3.16. This is surprising given that Theft or Robbery is typically a more stressful experience for customers but may be explained by the very small number of such claims (only 3 in the dataset).

Finally, when comparing accepted versus refused claims, accepted claims have an average satisfaction of 3.18, while refused has a score of 3.17, which suggests that the refusal decision itself does not dramatically impact satisfaction and indicates that customers who receive clear and timely communication about a refusal are able to accept the outcome.

Overall, the Satisfaction Analysis reveals that the customer satisfaction score is relatively uniform across most segments, hovering around 3.18, which suggests that the dissatisfaction drivers are process-wide rather than specific to any segment.

4.6. CONFORMANCE CHECKING

A comprehensive conformance checking analysis executed on the SafeDrive Insurance Claims Process reveals a striking 0% perfect conformance rate, demonstrating that real-world operations completely diverge from the intended design across all 50 distinct deviation patterns identified.

The most fragmented deviation across the workflow occurs within the Claim Decision activity, which is repeatedly flagged as an unexpected event (*Unexpected event*). This final decision emerges out of order across six independent deviation patterns: the main pattern affects 42.96% of cases (10K cases) with a throughput time (TPT) impact of +1 day, followed by secondary pattern A with 9.76% of occurrences (2.27K cases) and a delay of +2 days, secondary pattern B affecting 7.38% of instances (1.72K cases)

with a +3 days delay, and secondary pattern C appearing in 1.67% of lines (388 cases) adding +1 day. Furthermore, it manifests in two separate anomalous acceleration patterns of 1.65% (384 cases) and 1.53% (355 cases), where it incorrectly shortens the process with a TPT impact of -1 day in both instances, proving that final closure decisions are being made arbitrarily without sequence criteria, skipping mandatory technical validation steps.

In parallel, digital interaction channels suffer from severe workflow anomalies, with unexpected emails sent (Unexpected E-Mail Sent) splitting into five explicit deviations where the largest operational impact affects 24.35% of claims (5.67K cases), provoking a severe delay of +24 days, followed by secondary patterns at 5.51% of instances (1.28K cases) with a penalty of +23 days, 4.02% of occurrences (935 cases) adding +22 days, 0.96% of cases (224 cases) with a delay of +25 days, and 0.88% of situations (205 cases) introducing an additional +22 days to the process. This behavior is closely mirrored by unexpected emails received (Unexpected E-Mail received), which are distributed across five independent deviations where the primary pattern affects 22.97% of cases (5.35K cases), accumulating +21 days of retention, followed by secondary patterns affecting 5.39% of workflows (1.26K cases) generating +20 days, 4.27% of lines (994 cases) with an increase of +22 days, 0.87% of instances (202 cases) adding +22 days, and 0.86% of records (200 cases) introducing an additional +22 days. When these digital channels clog, the operation resorts to out-of-sequence manual telephone contacts via the Call: Request Information deviation, which surfaces at three distinct points: the first in 7.53% of cases (1.75K cases) adding a severe +42 days bottleneck, the second in 1.89% of cases (440 cases) with an impact of +36 days, and a third at 1.38% of instances (321 cases) introducing a +30 days delay.

The automated messaging infrastructure also reveals systemic instability manifesting in three distinct ways. First, through omission failures (*Missing event*), where the automated transmission of Automated SMS fails to occur across three scenarios, affecting respectively 17.30% of workflows (4.03K cases) with an impact of +9 days, 3.92% (912 cases) with +10 days, and 3.15% (734 cases) with +11 days. Second, redundant triggers (*Occurred too often*) of this automation occur in three separate patterns where 9.45% (2.2K cases) generate a +9 days delay, 2.11% (490 cases) bring a +10 days penalty, and 1.62% (376 cases) add +11 days. Third, the event triggers

entirely out of sequence (*Occurred out of sequence*) in 6.08% of instances (1.42K cases) with +9 days, in 1.65% of instances (384 cases) adding +11 days and in 1.34% of instances (312 cases) with +10 days.

The operational stages of technical assessment and appraisal present serious conformance issues divided between omissions and rework. The lack of technical closure (Assessment Closed as a *Missing event*) is identified across three process variants, led by a primary pattern of 13.56% with an impact of +30 days, accompanied by secondary deviations of 3.23% with +29 days and 2.62% with +23 days. Concurrently, essential scheduling and date activities are missing on a recurring basis, where Assessment Date disappears in deviations of 11.37% with +28 days, 2.63% with +25 days, and 1.97% with +31 days, while Assessment Scheduled is not executed in patterns of 11.22% with +30 days, 2.63% with +27 days, and 2.03% with +28 days. When these steps are actually performed, document intake failures trigger unnecessary repetitions (*Occurred too often*), where Assessment Scheduled repeats in 2.47% of instances (574 cases) adding +30 days, and Assessment Date duplicates in 2.45% of situations (571 cases), adding +28 days to the total cycle time.

Finally, these vulnerabilities culminate in severe misalignments during the financial settlement and closure phase. The complete omission of payment (Make Payment as a *Missing event*) occurs across three significant deviations, starting with a primary pattern of 23.25% (5.41K cases) with +20 days, followed by patterns of 5.20% (1.21K cases) with +20 days and 4.06% (944 cases) with +19 days and a fourth pattern of 0.92% (215 cases) with +16 days. When the payment does advance, it is executed out of the correct sequence (*Occurred out of sequence*) across three fronts led by 8.56% (1.99K cases) generating +20 days, and accompanied by secondary deviations of 2.24% (522 cases) with +19 days and 1.69% (393 cases) with +20 days, while also suffering from banking redundancy (*Occurred too often*) in 1.72% of situations (400 cases), inflating the TPT by an additional +20 days of rework.

To close the deviation cycle, the final closure of the claim (Close Claim) occurs entirely out of sequence in 2.31% of instances (538 cases), adding a logistical delay of +3 days. In conclusion, the analytical diagnosis exposes a highly reactive and unstandardized operational scenario, requiring SafeDrive Insurance to implement strict automated system blocks to halt final decisions before appraisal completion and

stabilize the alert infrastructure to completely eradicate these accumulated time inefficiencies. Additionally, a Cost Estimate was flagged as an unexpected event in 0.95% of cases (222 cases), introducing a severe time penalty of +42 days whenever it appears in the process.

While the 0% conformance rate reflects significant process deviations, it also raises questions regarding the suitability of the reference model. The idealized process may be overly restrictive, failing to account for legitimate operational variations that occur in practice.

4.7. TIME AND EFFICIENCY ANALYSIS

The Time and Efficiency Analysis isolates operational friction and quantifies process delays across the claim's lifecycle, exposing a clear operational divide between early-stage intake and late-stage administrative processing.

The front-end intake phase moves relatively quickly, requiring an average of just 8.40 days to transition a case from the initial incident to formal registration. In stark contrast, the back-end processing, investigation, and closure phase emerges as the primary organizational bottleneck, consuming an average of 24.39 days. Together, these phases drive the total end-to-end process to an average of 32.8 days. However, a look at the broader distribution reveals that the process median sits significantly lower at 15 days. This sharp statistical variance proves that the baseline average is heavily inflated by a small, problematic group of severe outlier cases, with maximum durations stretching up to 1.66K days.

When navigating the data deeper, massive operational discrepancies surface based on regional management and asset categorization. On a geographic level, Braga represents the most severe regional delay by a wide margin, saddled with a staggering average total processing time of 99.47 days.

This regional friction is closely mirrored by the physical assets involved in the claims. Commercial and standard passenger assets face prolonged administrative timelines, with Fleets and Light Vehicles experiencing near-identical operational friction, averaging 40.39 days and 39.87 days respectively. Conversely, two-wheeled transit routes appear much easier to process, allowing Motorcycles to clear the insurance workflow notably faster while maintaining a lower average of 29.31 days.

Beyond geographic and asset divisions, the underlying reason for a claim and its final adjudication decision heavily dictate the velocity of the workflow. Collision/Impact claims prove to be incredibly resource-intensive and procedurally complex, requiring an extensive average of 67.27 days to resolve. This lengthy lifecycle stands in sharp contrast to Theft or Robbery claims, which average 33.33 days, and Glass Breaking incidents, which represent the most streamlined, transactional workflow at just 23.06 days.

These timelines are further influenced by the final resolution of the claim itself. Claims that are ultimately Accepted require a lengthier verification, appraisal, and payout cycle that pushes their timeline to 32.85 days. Meanwhile, claims destined to be Refused are successfully flagged, audited, and filtered out of the system much faster within 24.81 days.

Ultimately, the data demonstrates that insurance processing times vary drastically across different operational segments, with the most severe volatility concentrated around specific claim reasons and regional districts. This extreme level of variance is a clear indicator of inconsistent handling procedures and fragmented resource distribution across regional offices. To mitigate these structural delays and elevate underperforming regions, there is a critical, immediate need for unified, cross-location workflow enforcement to standardize the claims journey from end to end.

4.8. CLAIMS REFUSAL ANALYSIS

The Claims Refusal Analysis dashboard examines the frequency and characteristics of refused claims across the company's portfolio. Of the 23,275 claims in the dataset, 155 were refused, resulting in an overall refusal rate of just 0.67%, making refusal a very rare outcome.

Despite this, the total value of refused claims amounts to just 14,568€ with an average refused claim of 93.99€, which is significantly lower than the overall average claim amount of 1,101.97€. This suggests that refused claims tend to involve lower financial amounts.

At the district level, Aveiro has the highest refusal rate by a clear margin, at 2.41%, sitting well above all other districts. Guarda follows with 0.87%, Faro with 0.44%, Lisboa with 0.35% and Coimbra with 0.25%. The rest of the districts have a refusal

rate of 0%. The particularly high refusal rate in Aveiro is a notable finding that with a further investigation, it may indicate inconsistent claim validation procedures, a higher prevalence of fraudulent submissions, or stricter local assessment standards in that region.

In terms of vehicle type, Light Vehicles have the highest refusal rate at 1.16%, followed by Motorcycles at 0.61% and Fleets at 0.48%. This is an interesting finding given that Motorcycles represent the largest volume segment, suggesting that the higher absolute number of motorcycle claims does not translate into a proportionally higher refusal rate.

Regarding claim reason, Glass Breaking has the highest refusal rate at 0.79%, followed by Collision/Impact at 0.23%. Notably, Theft or Robbery shows a refusal rate of 0.00%, which is surprising given that this category is typically associated with higher fraud risk. However, this result should be interpreted with caution given that only 3 Theft or Robbery claims exist in the entire dataset, making any conclusions statistically unreliable.

In terms of active processing time, refused claims are resolved faster at an average of 14.32 days compared to accepted claims at 24.46 days (which differs from the total end-to-end days analyzed previously). This suggests that refusals are identified and communicated relatively quickly, likely because the grounds for refusal are identified early in the assessment process without requiring the full investigation and payment workflow to be completed.

Finally, when comparing customer satisfaction between refused and accepted claims, the scores are virtually identical, where accepted claims have an average of 3.18 while refused claims an average of 3.17. This near-identical result is a positive finding, suggesting that the way SafeDrive Insurance communicates refusal decisions is generally well received by customers and does not significantly damage their overall experience.

Overall, while the refusal rate is very low at 0.67%, the geographic concentration of refusals in Aveiro and the higher refusal rate for Light Vehicles are patterns that deserve ongoing monitoring to detect potential fraud trends and ensure consistent claims assessment standards across all districts.

4.9. COMMUNICATION ANALYSIS

This section evaluates how SafeDrive Insurance manages its interactions with customers throughout the claim's lifecycle. The data reveals that the "Automated SMS" trigger is the most frequent activity across the entire workflow, registering 28,235 occurrences. However, despite this automation, there are significant gaps in proactive communication. A major red flag is that 11.6% of all claims (2,689 cases) involved a "Call: Request Information" event, generating a total of 4,651 calls. This clearly indicates that the automated updates are insufficient, forcing a large portion of the customer base to manually seek information, which directly explains the customer complaints about unclear communication.

A deeper geographical and demographic analysis of these manual calls reveals striking disparities. Even though Lisboa holds the vast majority of total claims, the districts of Aveiro and Coimbra are driving the most manual inquiries, with 790 and 636 cases with calls respectively.

To contextualize these metrics, a regional inquiry rate was analyzed by dividing the number of distinct cases with information requests by the total claim volume of each region, extracted from the Business Analysis dashboard. This cross-analysis reveals a profound operational asymmetry between branches: while Lisboa demonstrates high efficiency with a manual inquiry rate of just 1.51% (214 calls out of 14,131 claims), Aveiro escalates to 19.46% (790 calls out of 4,059 claims), Coimbra reaches 39.14% (636 calls out of 1,625 claims), and Beja records the highest administrative friction in the country with 41.88% of its policyholders (325 calls out of 776 claims) forced to manually pull status updates. This proportional reality points to localized process bottlenecks or differences in staff diligence across regional offices.

Additionally, this communication breakdown disproportionately affects the Motorcycle segment, which accounts for a massive 62.07% of all cases requiring a manual call. Predictably, this need to constantly follow up negatively impacts the customer experience, as cases involving information requests generally present lower satisfaction scores compared to the baseline.

Furthermore, a significant imbalance exists in email correspondence: there were 22,607 instances of "E-Mail Received" compared to only 16,489 instances of "E-Mail

Sent". This metric strongly suggests that customers are reaching out to the company more frequently than they receive updates, serving as another definitive sign that proactive communication is failing. This lack of transparency likely drives dissatisfaction, aligning perfectly with the high volume of information requests coming from the Aveiro and Coimbra districts and the Motorcycle segment.

5. CONCLUSIONS

5.1. SUMMARY OF FINDINGS

The process mining initiative at SafeDrive Insurance revealed a highly complex operational reality, highlighted by the existence of over 5,400 distinct process variants and a striking 0% conformance rate to the ideal model. Furthermore, average customer satisfaction remains moderate at a score of 3.18, suggesting that customers are not fully satisfied with the current service. Business volume is predominantly driven by the Lisboa district and Motorcycle claims, while processing times vary considerably across different districts and claim categories.

5.2. MAIN PROBLEMS IDENTIFIED

The current workflow is affected by several critical inefficiencies. Firstly, the existence of 5.45K variants indicates a severe lack of process standardization. This complexity contributes to significant geographic inequality, with the Braga district requiring an average of 99 days to process a claim, closely followed by Porto (96 days) and Guarda (89 days), compared to a national average of approximately 32.8 days. Similarly, there is a significant difference in processing times for different types of claims: Collision/Impact cases take about 67 days to resolve, whereas Glass Breaking incidents require only 23 days. Procedural deviations further exacerbate these processing delays.

The omission of the "Assessment Closed" step in 13.56% of cases (3,160 instances) adds a significant delay of 30 days to the workflow, while unplanned events, such as unexpected emails sent and received, add average delays of 24 and 21 days respectively when they occur out of sequence.

Finally, there is a significant communication failure within the company, evidenced by the 11.6% of customers (2,689 cases) who were forced to call to request information, highlighting a clear failure in proactive communication.

5.3. RECOMMENDATIONS

To address these systemic issues, SafeDrive Insurance must implement targeted operational changes.

Firstly, the company should standardize the process by reducing the current 5,450 variants into a much smaller, strictly enforced set of approved operational paths. At the same time, management must focus on Braga, Porto, and Guarda to immediately investigate why these specific districts take nearly 4 times longer than the national average.

To address the severe delays in specific segments, the creation of a dedicated fast-track process for highly complex Collision/Impact claims is strongly advised.

Additionally, SafeDrive Insurance needs to overhaul its communication strategy by sending automatic updates at each critical milestone to ensure proactive communication and reduce the need for customers to call in for status updates.

At a system level, it is crucial to enforce system safeguards to guarantee that the "Assessment Closed" step is completed, as the frequent omission of this milestone artificially inflates throughput time by up to 30 days.

Finally, providing comprehensive staff training will be essential to reduce the high frequency of unexpected activities and ensure strict adherence to standard procedures.

5.4. VALUE GENERATED

The application of process mining techniques successfully revealed hidden operational inefficiencies that were previously invisible to management.

This project provides clear, data-backed guidance on where to focus continuous improvement efforts, specifically highlighting bottlenecks in Braga, Porto and Guarda, Collision/Impact claims process, and the Assessment phase.

By quantifying the exact time penalty of each process deviation, the analysis establishes a strong business case for immediate investment in process standardization and advanced automation.